

Proposal for Targeted Signal Search at 1.81 GHz in ADMX Data: Gravitational Wave Signatures from Vacuum Phase Transitions

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Abstract

I propose a specific search for narrow-band electromagnetic signals centered at **1.811 GHz** within existing ADMX L-band datasets. This frequency is predicted by the Axiomatic Physical Homeostasis (APH) framework as the spectral signature of a vacuum topological filtration event occurring at the GUT scale ($E \approx 2 \times 10^{16}$ GeV). Unlike standard axion models, this signal arises from high-frequency tensor perturbations converting to photons via the Inverse Gertsenshtein Effect. I provide the theoretical derivation of the frequency, the expected fine structure (± 21 MHz sidebands), and a computational verification method utilizing the Universal Thin Filament Code (uTFC).

1 Introduction

The current search for Dark Matter is largely focused on the QCD axion, motivated by the strong CP problem. However, the sensitivity of axion haloscopes like ADMX extends to any weak electromagnetic conversion process involving a background magnetic field.

I present a theoretical framework that predicts a specific, discrete resonance in the stochastic gravitational wave background (SGWB). This signal originates not from inflation, but from the relaxation of the vacuum geometry from a primordial Sedenion phase ($\mathbb{S}$) to the stable Octonionic phase ($\mathbb{O}$) characteristic of the Standard Model. This transition acts as a *geometric stiffness* filter, generating a residual phonon background in the vacuum metric.

2 Theoretical Derivation of the Signal

2.1 The Geometric Stiffness Parameter

The fundamental parameter of the theory is the geometric stiffness of the vacuum, β . Stability analysis of the non-associative bulk geometry yields a critical value for the stability of the QCD vacuum:

$$\beta_{QCD} = \frac{\text{Dim}(\text{Bulk})}{\text{Measure}(\text{Cycle})} = \frac{6}{\pi} \approx 1.90986 \quad (1)$$

This super-linear stiffness ($\beta > 1$) is responsible for the mass gap and confinement. It also dictates the relaxation frequency of the vacuum geometry.

2.2 The 1.811 GHz Resonance (“The Sedenion Scream”)

The fundamental frequency of this relaxation, ν_3 , is derived from the entropic capacity of the stable 3-cycles in the vacuum filtration. This spectrum is geometrically anchored to the 21-cm Hydrogen hyperfine transition ($\nu_H \approx 1.4204$ GHz) by the projection factor of the $SU(2)$ root length, $\sqrt{2}$:

$$\nu_0 \equiv \frac{\nu_H}{\sqrt{2}} \approx 1.0044 \text{ GHz} \quad (2)$$

The primary resonance corresponding to the S^3 fiber decay is given by:

$$\nu_3 = \nu_0 \cdot \frac{3}{2}\zeta(3) \approx 1.0044 \cdot 1.803 \approx \mathbf{1.811 \text{ GHz}} \quad (3)$$

2.3 Fine Structure and Sidebands

The model predicts specific modulation of this carrier frequency due to the relaxation of the embedding dimensions ($SO(7)$). This manifests as sidebands at:

$$\Delta f \approx \pm 21 \text{ MHz} \quad (4)$$

The detection of a peak at 1.811 GHz accompanied by these specific sidebands would constitute a high-confidence signature of this phenomenology, distinct from terrestrial radio-frequency interference (RFI).

3 Computational Verification: The uTFC Framework

The theoretical derivation is supported by numerical simulations using the Universal Thin Filament Code (uTFC), originally developed for magnetospheric plasma stability.

3.1 The Wolf Parameterization Mapping

I map the discrete micro-states of the vacuum (modeled as a cellular automaton, Rule 30) to continuous pressure $P(s)$ and density $\rho(s)$ profiles using the Wolf Parameterization. The stability of the vacuum is determined by the Sturm-Liouville eigenvalue equation:

$$\frac{d}{ds} \left(P(s) \frac{d\xi}{ds} \right) + \rho(s) \omega^2 \xi = 0 \quad (5)$$

Numerical solutions confirm that a stable, real eigenmode spectrum emerges only when the stiffness parameter approaches the critical value $\beta \approx 1.91$. The fundamental eigenfrequency of this stable branch scales directly to the predicted 1.8 GHz observable when renormalized to the electroweak scale.

4 Proposed Detection Strategy

4.1 The Inverse Gertsenshtein Effect

The detection mechanism relies on the conversion of tensor gravitational waves ($h_{\mu\nu}$) into photons (A_μ) in the presence of a static magnetic field (B_0). The effective interaction Lagrangian is:

$$\mathcal{L}_{int} \propto \frac{1}{M_{Pl}} h_{\mu\nu} F^{\mu\rho} F^\nu_\rho \quad (6)$$

This allows ADMX cavities, tuned to 1.811 GHz, to function as gravitational wave receivers.

4.2 Discrimination Protocol

To distinguish this signal from scalar axions or noise, I propose analyzing the data for **Tensor Polarization Modulation**. Unlike a scalar source, the coupling efficiency of a tensor wave depends on the angle θ between the cavity axis and the incident wave vector. As the Earth rotates, a cosmic tensor signal will exhibit a characteristic sidereal modulation distinct from the axion haloscope signal.

5 Conclusion and Request

I respectfully request that the ADMX collaboration review existing L-band data for power excesses in the **1.811 GHz $\pm$ 21 MHz** window.

The observation of such a signal would provide direct evidence for the geometric stabilization of the vacuum at the GUT scale. I'm prepared to provide the full Python code for the uTFC simulation to assist in signal template generation.

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